

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>
A change in temperature of 30 °C in 1 liter of water , requires $1\text{ cal/g}^{\circ}\text{C} * (1\text{ kg}) * (1000\text{ g/l}) * 30^{\circ}\text{C} = 30,000\text{ calories} = 30\text{ Kcal}.$
A change in temperature of 30 °C in 1 kg of aluminum , requires $0.215\text{ cal/g}^{\circ}\text{C} * (1\text{ kg}) * (1000\text{ g}) * 30^{\circ}\text{C} = 6.45\text{ Kcal}.$
</think>
<answer> G . 30 Kcal , 6 . 45 Kcal </answer>